



BACHELOR OF INFORMATION AND COMMUNICATIONS TECHNOLOGY IN INTERNET OF THINGS

PROJECT DESCRIPTION AND SPECIFICATION DOCUMENT

DATA MANAGEMENT (DTMG202) – Mrs E. Naicker

WEB DEVELOPMENT (WBDV202) – Mr P. Monyeki

DUE DATE: 29 September 2026

2026

Instructions to Students

- Students should work in groups of 6. All students are to take personal responsibility for the quality of their work. (There might be instances where students are doing only one of the modules (DTMG202 and WBDV202), but they can be still be part of the group)
- You are required to develop a web application that will provide a real-world solution that could improve the function of the business. You are required to create an interactive database using PostgreSQL or SQL Server relational database management system (RDBMS). Populate the database, write interactive queries and modifications on the database, and explore other aspects such as constraints.
- DUT does not tolerate academic dishonesty. Students are warned against copying from the Internet, the unethical use of large-language models and chatbots such as ChatGPT, as well as other forms of plagiarism.
 - In accordance with the [DUT Plagiarism Policy](#), if a student or group is found guilty of plagiarism, they will **be deducted 50% of their project mark immediately** and be referred to the Programme Coordinator and Head of Department for further disciplinary action.
- The project submission will consist of two deliverables:
 - Project presentation.
 - System demonstration.

Embedded knowledge and skills required by students

As this project is integrated between three modules (**Data Management and Web Development**), the embedded knowledge and skills required by students overlaps the learning outcomes of each of these modules.

Data Management

Students should understand data storage, analysis and processing of data. An understanding of databases must be demonstrated by:

1. Determining the data requirements (business rules) of the web application.
2. Describing a visual representation of data organization by drawing a Entity Relationship Diagram (ERD) (include cardinalities, attributes, relationships).
3. Describing the implementation (population and storage of data in database) and protection of the database.

Web Development (wbdv202)

Each group must design and develop a website for a small business or organisation. The website must provide information about the business, allow users to interact with the system, and demonstrate the use of front-end and basic back-end technologies. An understanding of creating web applications must be demonstrated by:

1. Describing the basic knowledge of HTML, CSS, and JS.
2. Mastering the design of web applications.
3. Describing how to create websites and web forms.
4. Verification & authentication
5. Functionality such as login, searching etc.
6. User friendly interface
7. Web page linked to the dataset

This project also develops the DUT Graduate Attributes, by:

- Allowing students to work in self-managed groups, enabling them to work independently and collaboratively.
- Developing knowledge practitioners who are able to apply theoretical knowledge to practical systems.

Resources required

- A computer with access to the Internet and appropriate software for development.
- Access to the DUT library and DUT Moodle Learning Management System.
- Access to DUT IoT Project Workshop Laboratories.

Description of the Project

You are required to develop a web application that will provide a real-world solution that could improve the function of the business. You are required to create an interactive database using PostgreSQL or SQL Server relational database management system (RDBMS). Populate the database, write interactive queries and modifications on the database, and explore other aspects such as constraints.

Subject lecturers will be responsible for providing any technical advice to students related to the subject matter from their module. *Each group is expected to nominate a group leader, who will be the key contact between the group and staff mentor.*

Deliverable Structure

Students are responsible for ensuring that they prepare their deliverables ahead of time; including rehearsing the timing of their presentations and arranging backup/spare hardware for their demonstrations. If a group does not deliver their presentation/demonstrate their system at the allotted deadline, any opportunity to resubmit (at the discretion of the staff) will result in a mark that is **capped at 50%**.

Marking Scheme

Module	Assessment criteria	Mark Breakdown					Total Mark for Module
		Poor	Below Average	Average	Good	Very Good	
Data Management	Description of Systems and their Functions/Processes.	0	1	2	3	4	/25
	Description of data requirements.	0	1	2	3	4	
	Visual representation of system data, relationships, structures and constraints.	0	1	2	3	4	
	Database in third normal form	0	0	0.5	1	1	
	Demonstration of queries within the database	0	1-2	3	4-5	6	
	Presentation of Document	0	0	1	2	3	
	Communication Skills	0	0	1	2	3	
Web Development	Html structure and content	0	1	2	3	4	/25
	CSS: Clear visual representation of the system design	0	1	2	3	4	
	Description of Systems and their major Functions/Processes.	0-1	2-3	4-5	6-7	8-9	
	Alignment between mock-up views and the database	0	1	2	3	4	
	Communication Skills	0	1	2	3	4	

In the event of any of the following, the group's mark per the scheme above will be scaled down by a penalty factor of between 10% to 50%, depending on the severity of the transgression. This will be determined at the discretion of the academic mentor.

- Failure to attend progress meetings with academic mentor.

- Failure to attend booked laboratory sessions.
- Failure to obey reasonable directives from academic mentor related to improving the quality of the project.
- Failure to cooperate with group members.
- Failure to submit any deliverables (Materials List, etc.) to mentor by the stipulated due dates.

Project Document should include the following:

- Description of system
- Description of each functionality of system.
- Your data requirements /business rules.
- ERD
- Normalisation

Project document must be uploaded onto Moodle on **25 September 2026 by 16h00**

Hardcopy of document must be handed during presentation of project on **28 September 2026**